

Check fitted flocker model

Specify model to check:

```
spec <- params$spec #final_species_eco_sorted[6] # 1
spec
```

```
[1] "Ruby-throated Hummingbird"
```

```
buffer_km <- params$buffer_km
buffer_km
```

```
[1] 750
```

```
fold <- params$fold
# fold

results_dir <- params$results_dir
results_dir
```

```
[1] "M:/Documents/DEBTs/analysis/Schifferle_BBS_occupancy_models_2023/results/full_model"
```

Load fitted model:

```
load(file = file.path(results_dir,
                       paste0("out_", spec, "_fm_buffer", buffer_km, ".RData"))) # xx adjust
#print(out)

posterior <- brms::as_draws(out)
```

Check whether MCMC worked:

Do we have divergent transitions?

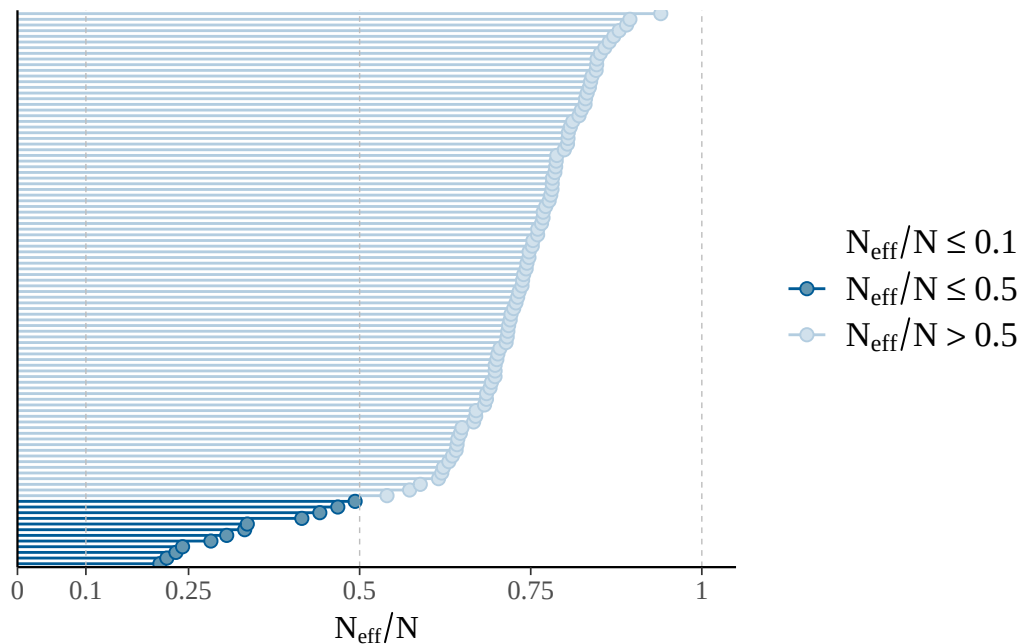
```
[1] "divergent transitions: 0"
```

Explorations regarding divergent transitions:

...

Is effective number of MCMC samples large enough:

The larger the ratio of n_{eff} to N (number of samples), the better.



```
[1] "effective sample size fine"
```

```
[1] "n_eff ratio bulk ESS:"
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.2081	0.7652	0.9332	0.9985	1.1964	2.5496

```
[1] "n_eff ratio tail ESS:"
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.3467	0.6688	0.7415	0.7262	0.8102	0.9400

```
[1] "effective sample size in bulk and tail fine"
```

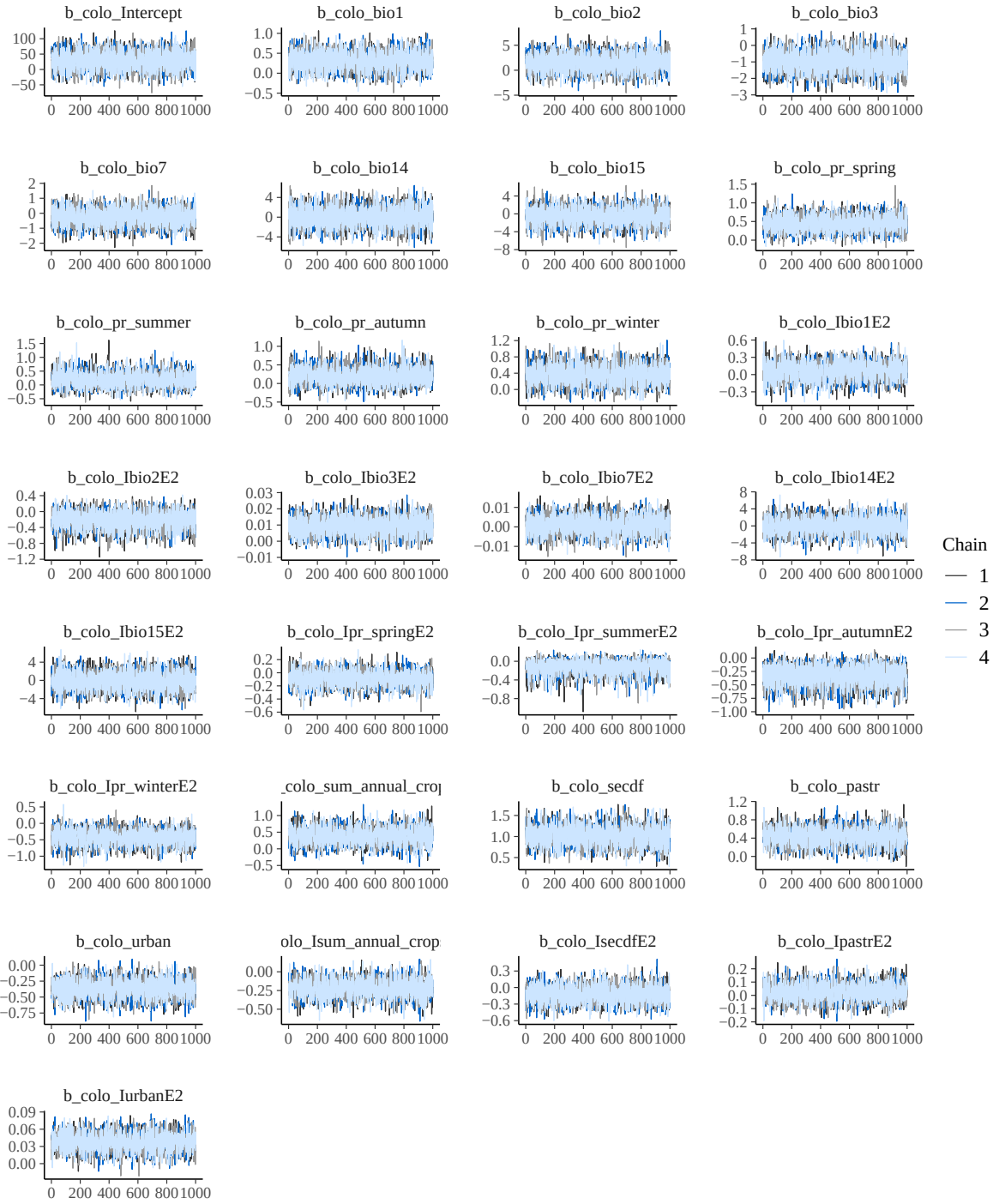
Do chains mix well:

No divergences to plot.

No divergences to plot.



No divergences to plot.





Are R-hat values fine:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9992	0.9996	1.0000	1.0001	1.0004	1.0026

```
[1] "All R-hat values < 1.01"
```

Print model posteriors

```
[1] "parameters for which uncertainty interval doesn't overlap zero:"
```

```
[1] "b_occ_bio14_3yrs"          "b_occ_lbio14_3yrsE2"
```

```
[3] "b_occ_lpr_winter_3yrsE2"  "b_occ_secdf_3yrs"
```

```
[5] "b_occ_lsecdf_3yrsE2"      "b_occ_lpastr_3yrsE2"
```

```
[7] "b_occ_lurban_3yrsE2"
```

```
[1] "b_colo_pr_spring"          "b_colo_lpr_autumnE2" "b_colo_secdf"
```

```
[4] "b_colo_pastr"             "b_colo_urban"         "b_colo_lurbanE2"
```

```
[1] "b_ex_bio1"                "b_ex_sum_annual_crops" "b_ex_pastr"
```

```
[4] "b_ex_urban"               "b_ex_lsecdfE2"         "b_ex_lpastrE2"
```

```
[7] "b_ex_lurbanE2"
```

